

Rational equations

Clear the denominators, solve, then throw away the roots the original equation never allowed.

LESSON 28–29

2 × 40 MIN

ALGEBRA 10, §2.1

P1 · 1.7 (EXTENSION)

LESSON CLOCK

7'

14'

14'

16'

20'

9'

Why a check is compulsory here

7 min

The method

14 min

Extraneous roots

14 min

Disguised quadratics

16 min

Practice

20 min

Homework

9 min

LEARNING OBJECTIVES

- State the domain of a rational equation before solving it.
- Solve an equation by finding the lowest common denominator.
- Identify and reject extraneous roots.
- Solve equations that become quadratic after a substitution.

TEXTBOOK

National

pp. 94–103

Cambridge

Pure Mathematics 1, pp. 18–23

Workbook

P1 Exercise 1F

01

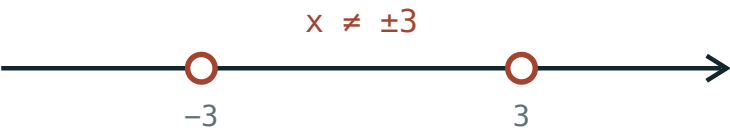
Explanation

The domain comes first

WRITE IT BEFORE YOU SOLVE

Every denominator gives a forbidden value. Write $x \neq \dots$ on the first line, before any algebra. It is not a formality: it is the only thing that will tell you which of your final answers is real.

For $\frac{3}{x-2} + \frac{1}{x} = 2$ the domain is $x \neq 0$ and $x \neq 2$.



The two excluded points, marked as open circles before any solving begins.

The method

1. State the domain.
2. Factorise every denominator, and find the lowest common denominator.
3. Multiply every term by the LCD, and cancel.
4. Solve the polynomial equation that results.
5. Reject any root that is outside the domain.

$$\frac{3}{x-2} + \frac{1}{x} = 2 \Rightarrow 3x + (x-2) = 2x(x-2)$$

$$4x - 2 = 2x^2 - 4x \Rightarrow 2x^2 - 8x + 2 = 0 \Rightarrow x^2 - 4x + 1 = 0 \Rightarrow x = 2 \pm \sqrt{3}$$

Both roots are allowed, so both are answers.

MULTIPLY EVERY TERM

A term with no fraction is multiplied by the LCD too. Forgetting the right-hand side is the commonest arithmetic error in the topic.

Extraneous roots — where they come from

Multiplying by $x - 3$ is only reversible when $x \neq 3$. If it happens to be zero, you have multiplied both sides by zero, and the new equation is **not equivalent** to the old one — it has extra solutions the original never had.

$$\frac{x^2}{x-3} = \frac{9}{x-3} \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$$

But $x = 3$ is outside the domain. The only solution is $x = -3$.

THE CHECK IS PART OF THE ANSWER

Never write a final answer without testing it against the domain line you wrote at the start. Examiners award a mark for the rejection, and take one for its absence.

Disguised quadratics

When the same expression appears twice, name it and the equation becomes quadratic:

$$x^2 + \frac{1}{x^2} = 2.5, \text{ let } t = x + \frac{1}{x}$$

Then $t^2 = x^2 + 2 + \frac{1}{x^2}$, so the equation reads $t^2 - 2 = 2.5$, giving $t = \pm 1.5$ — two much easier equations to finish.

The other standard shape is $\frac{x}{x+1} + \frac{x+1}{x} = 2.5$: set $t = \frac{x}{x+1}$ and it becomes $t + \frac{1}{t} = 2.5$.

02 Worked examples

EXAMPLE 1 Solve $\frac{2}{x} + \frac{3}{x+1} = 2$.

① Domain $x \neq 0, -1$.

② LCD $x(x+1)$.

$$2(x+1) + 3x = 2x(x+1)$$

③ $5x + 2 = 2x^2 + 2x$

④ $2x^2 - 3x - 2 = 0 \Rightarrow (2x+1)(x-2) = 0$

Both allowed.

ANSWER

$$x = 2 \text{ or } x = -0.5$$

EXAMPLE 2 Solve $\frac{x}{x-1} - \frac{1}{x-1} = 3$.

① Domain $x \neq 1$.

② Same denominator: $\frac{x-1}{x-1} = 1$.

For every allowed x .

③ $1 = 3$

A contradiction.

ANSWER

No solution

EXAMPLE 3 Solve $\frac{x^2}{x-2} = \frac{4}{x-2}$.

① Domain $x \neq 2$.

② $x^2 = 4 \Rightarrow x = \pm 2$

③ $x = 2$ is excluded.

ANSWER

$x = -2$ only

03 Interactive model

Show the graph of each side and let the class see where the removed point sits.

Domain first

Domain of $\frac{1}{x-5} = 2$:

$x \neq 0$

$x \neq 5$

$x > 5$

$\mathbb{R}$

Solving $\frac{x^2}{x-3} = \frac{9}{x-3}$ gives:

$x = \pm 3$

$x = 3$

$x = -3$

no solution

An extraneous root appears because:

of arithmetic error

we multiplied by something that can be zero

the equation is wrong

it never does

Solve $\frac{2}{x} = 1$:

$x = 2$

$x = 0.5$

$x = 0$

no solution

For $\frac{1}{x} + \frac{1}{x-1} = 5$ the LCD is:

x

$x - 1$

$x(x - 1)$

x^2

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Rational equation	Ratsional tenglama	Рациональное уравнение
Denominator	Maxraj	Знаменатель
Lowest common denominator	Eng kichik umumiy maxraj	Наименьший общий знаменатель
Extraneous root	Chet ildiz	Посторонний корень
Domain restriction	Aniqlanish sohasi cheklovi	Ограничение области определения
Clearing fractions	Kasrlardan qutulish	Освобождение от дробей
Equivalent equations	Teng kuchli tenglamalar	Равносильные уравнения
Check (substitution)	Tekshirish	Проверка
Substitution variable	Yordamchi o'zgaruvchi	Вспомогательная переменная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first line of a rational equation should be:

the LCD

the domain

the answer

a graph

Multiplying by the LCD may introduce:

lost roots

extraneous roots

nothing

fractions

$x - 1$ $x - 1$ equals 1:

always

for $x \neq 1$

never

only at $x = 1$

For $x^2 + \underline{1} x^2 = 2.5$ a good substitution is:

$$t = x^2$$

$$t = x + \frac{1}{x}$$

$$t = x - 1$$

none

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $\frac{1}{x-4} = 3$

ANSWER $x \neq 4$

2. Solve $\frac{6}{x} = 2$

ANSWER $x = 3$

3. Solve $\frac{1}{x+1} = 1$

ANSWER $x = 0$

4. Solve $\frac{x}{2} = \frac{3}{x}$

ANSWER $x = \pm\sqrt{6}$

5. Domain of $\frac{1}{x} + \frac{1}{x-2}$

ANSWER $x \neq 0, 2$

6. LCD of $\frac{1}{x}$ and $\frac{1}{x+3}$

ANSWER $x(x+3)$

7. Solve $\frac{4}{x-1} = 2$

ANSWER $x = 3$

Medium · the standard question

7 PROBLEMS

1. Solve $\frac{2}{x} + \frac{3}{x+1} = 2$

ANSWER $x = 2, -0.5$

2. Solve $\frac{x}{x+2} = \frac{3}{x}$

ANSWER $x = 6$ or $x = -1$

3. Solve $\frac{1}{x-1} + \frac{1}{x+1} = 1$

ANSWER $x = 1 \pm \sqrt{3}$

4. Solve $\frac{x^2}{x-2} = \frac{4}{x-2}$

ANSWER $x = -2$

5. Solve $\frac{3}{x} - \frac{2}{x^2} = 1$

ANSWER $x = 1, 2$

6. Solve $\frac{x+1}{x-1} = \frac{x-1}{x+1}$

ANSWER $x = 0$

7. Solve $\frac{x}{x-3} - \frac{3}{x-3} = 2$

ANSWER no solution

Hard · stretch and reason

7 PROBLEMS

1. Solve $x^2 + \frac{1}{x^2} = 2.5$

ANSWER $x = \pm 2, \pm 0.5$

2. Solve $\frac{x}{x+1} + \frac{x+1}{x} = 2.5$

ANSWER $x = 1, -2$

3. Solve $\frac{1}{x^2-1} + \frac{1}{x-1} = 1$

ANSWER $x = 2$ or $x = -1$ rejected

4. Solve $\frac{2x}{x^2-4} + \frac{1}{x-2} = \frac{3}{x+2}$

ANSWER $x = -8$

5. Find k so $\frac{x}{x-2} = k$ has no solution

ANSWER $k = 1$

6. Solve $\frac{x^2-4}{x^2-2x} = 1$

ANSWER no solution

7. Solve $(x + \frac{1}{x})^2 - 4(x + \frac{1}{x}) + 3 = 0$

ANSWER $x = 1$ (from $t = 3: \frac{3 \pm \sqrt{5}}{2}$)

07 Homework

Homework — 5 tasks

Every solution must open with the domain and close with the check.

1. Solve $\frac{3}{x} + \frac{2}{x-1} = 3$.
2. Solve $\frac{x}{x-4} = \frac{2}{x+1}$.
3. Solve $\frac{x^2}{x+3} = \frac{9}{x+3}$ and say which root is rejected and why.
4. Solve $\frac{1}{x} + \frac{1}{x+2} = \frac{3}{4}$.
5. Solve $x^2 + \frac{4}{x^2} = 5$ using a substitution.

Systems of rational equations

Two rational equations at once — substitution, symmetric sums, and the domain that must survive both.

LESSON 30–31

2 × 40 MIN

ALGEBRA 10, §2.2

P1 · 1.7

LESSON CLOCK

7'

18'

19'

14'

18'

4'

What is new	7 min
Substitution	18 min
Symmetric systems	19 min
Graphical meaning	14 min
Practice	18 min
Homework	4 min

LEARNING OBJECTIVES

- Solve a system containing rational equations by substitution.
- Use the sum-and-product substitution for symmetric systems.
- State the combined domain of a system.
- Interpret the solutions as points of intersection.

TEXTBOOK

National	pp. 104–113
Cambridge	Pure Mathematics 1, pp. 18–23
Workbook	P1 Exercise 1F

01

Explanation

The combined domain

BOTH EQUATIONS MUST ACCEPT THE ANSWER

A pair (x, y) solves the system only if it lies in the domain of **both** equations. So the excluded values of each are excluded from the whole system.

Everything else is the ordinary technique of Grade 9 — eliminate a variable, solve, substitute back — with the extra step of clearing denominators first.

Substitution

When one equation gives a variable cheaply, use it:

$$x + y = 5 \text{ and } \frac{1}{x} + \frac{1}{y} = \frac{5}{6}$$

The second equation is $\frac{x+y}{xy} = \frac{5}{6}$, so $\frac{5}{xy} = \frac{5}{6}$ and $xy = 6$. With $x + y = 5$ the numbers are the roots of $t^2 - 5t + 6 = 0$.

$(x, y) = (2, 3) \text{ or } (3, 2)$

TWO ORDERED PAIRS, NOT ONE

A symmetric system almost always gives both orders. Writing only one loses half the answer.

Symmetric systems

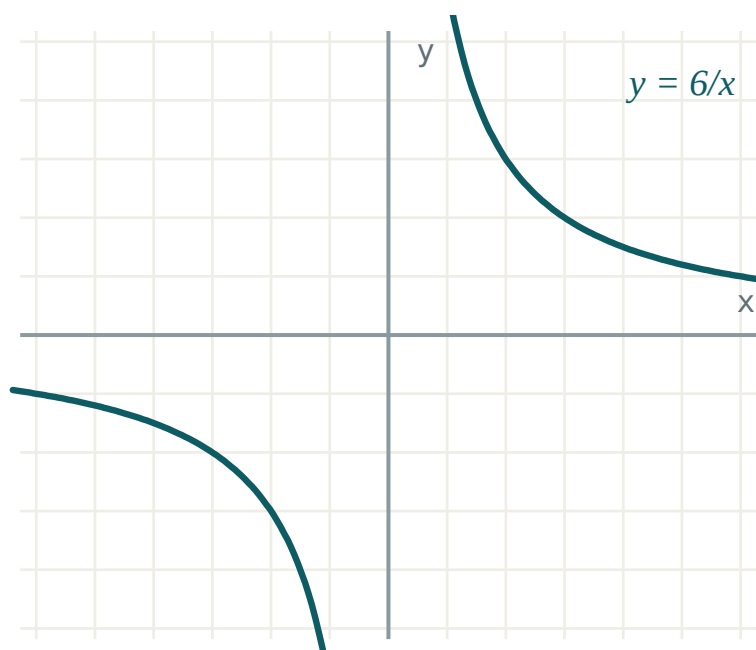
A system is **symmetric** when swapping x and y leaves it unchanged. Every such system can be rewritten in $s = x + y$ and $p = xy$:

EXPRESSION	IN s AND p
$x^2 + y^2$	$s^2 - 2p$
$\frac{1}{x} + \frac{1}{y}$	$\frac{s}{p}$
$x^3 + y^3$	$s^3 - 3ps$
$(x - y)^2$	$s^2 - 4p$

Solve for s and p , then recover x and y as the roots of $t^2 - st + p = 0$.

What the solutions mean

Each equation is a curve. The solutions of the system are the points where the curves cross — so the number of solutions is the number of intersections.



$xy = 6$ is a hyperbola; $x + y = 5$ is a line. Two crossings, two solutions.

A system with no solution is a line missing its curve entirely; a system with one solution is a tangency.

02 Worked examples

EXAMPLE 1 Solve $x + y = 7$, $\frac{1}{x} + \frac{1}{y} = \frac{7}{12}$.

$$\textcircled{1} \quad \frac{x+y}{xy} = \frac{7}{12}$$

$$\textcircled{2} \quad \frac{7}{xy} = \frac{7}{12} \Rightarrow xy = 12$$

$$\textcircled{3} \quad t^2 - 7t + 12 = 0$$

$$\textcircled{4} \quad t = 3, 4$$

ANSWER

$(3, 4)$ and $(4, 3)$

EXAMPLE 2 Solve $xy = 12$, $x^2 + y^2 = 25$.

① $s^2 - 2p = 25$ with $p = 12$.
 $s^2 = 49$

② $s = \pm 7$

③ $s = 7: t^2 - 7t + 12 = 0 \Rightarrow 3, 4.$

④ $s = -7: t^2 + 7t + 12 = 0 \Rightarrow -3, -4.$

ANSWER $(3, 4), (4, 3), (-3, -4), (-4, -3)$ **EXAMPLE 3** Solve $\frac{2}{x} - \frac{1}{y} = 1$, $\frac{1}{x} + \frac{2}{y} = 3$.

① Let $u = \frac{1}{x}$, $v = \frac{1}{y}$.
 $2u - v = 1, u + 2v = 3$

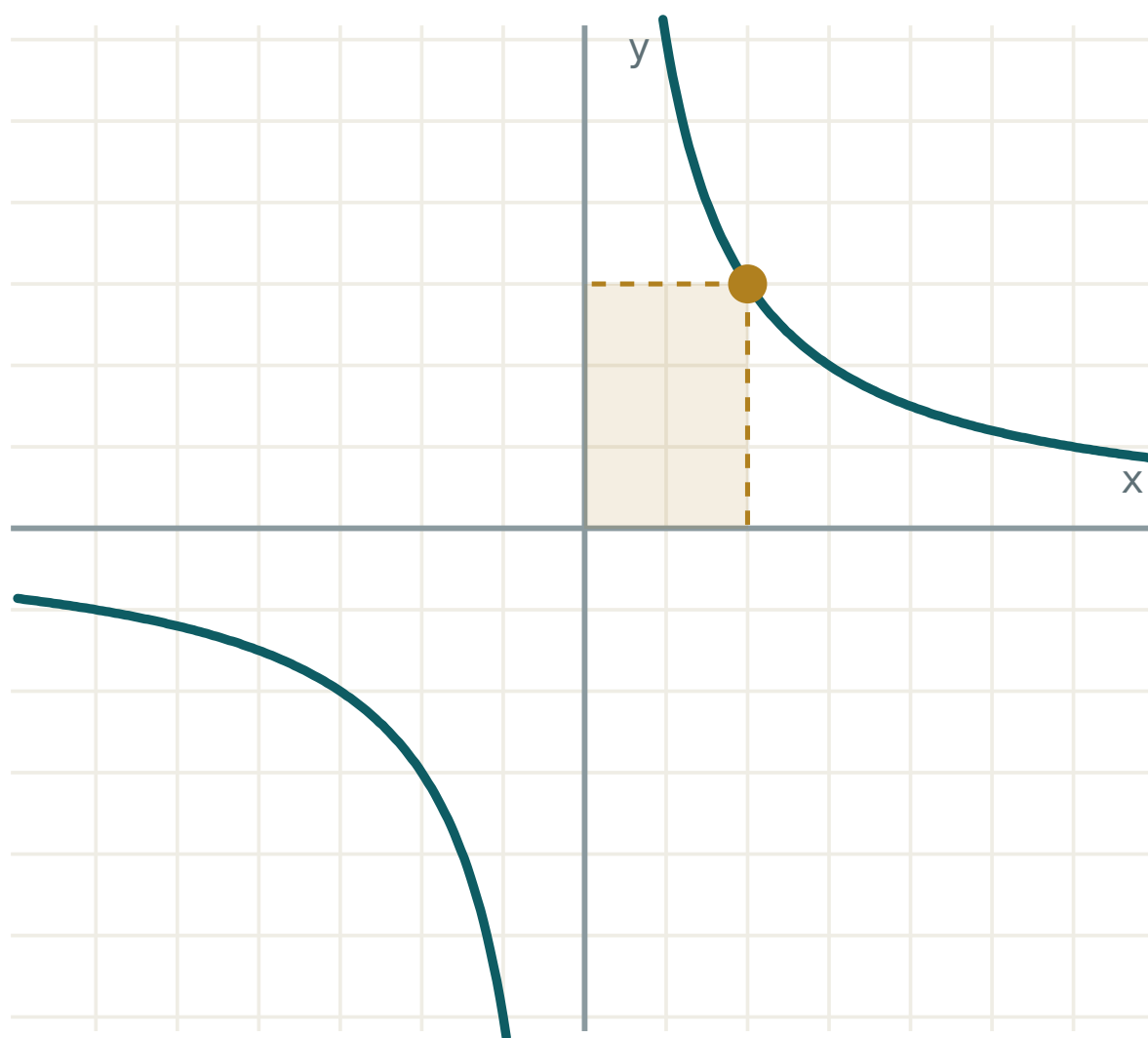
② Solve: $u = 1, v = 1.$

③ $x = 1, y = 1$

ANSWER $(1, 1)$ **03 Interactive model**

Plot both curves and count the crossings before solving.

The graph of $y = k/x$


 k 6

 x 2

 $y = k/x$ 3

 $x \cdot y$ 6

quadrants I and III

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
System of equations	Tenglamalar sistemasi	Система уравнений
Substitution method	O'rniga qo'yish usuli	Метод подстановки
Symmetric system	Simmetrik sistema	Симметричная система
Sum and product	Yig'indi va ko'paytma	Сумма и произведение
Combined domain	Umumiy aniqlanish sohasi	Общая область определения
Point of intersection	Kesishish nuqtasi	Точка пересечения
Consistent system	Birgalikdagi sistema	Совместная система
Inconsistent system	Birgalikda bo'lmagan sistema	Несовместная система

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A symmetric system usually gives:

one pair

two ordered pairs

no pairs

infinitely many

$x^2 + y^2$ in terms of s, p is:

$s^2 + 2p$

$s^2 - 2p$

$s^2 - p$

$2s - p$

For $\frac{1}{x} + \frac{1}{y}$ a good substitution is:

$u = x, v = y$

$u = \frac{1}{x}, v = \frac{1}{y}$

$u = xy$

none

The solutions of a system are:

the roots of one equation

the points where the curves meet

the domains

the asymptotes

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $x + y = 5$, $xy = 6$

ANSWER $(2,3), (3,2)$

2. Solve $x + y = 8$, $xy = 15$

ANSWER $(3,5), (5,3)$

3. Solve $x - y = 1$, $xy = 6$

ANSWER $(3,2), (-2,-3)$

4. Write $x^2 + y^2$ with s, p

ANSWER $s^2 - 2p$

5. Write $\frac{1}{x} + \frac{1}{y}$ with s, p

ANSWER $\frac{s}{p}$

6. Domain of $\frac{1}{x} + \frac{1}{y} = 1$

ANSWER $x \neq 0, y \neq 0$

7. Solve $xy = 4$, $x = y$

ANSWER $(2,2), (-2,-2)$

Medium · the standard question

7 PROBLEMS

1. Solve $x + y = 7$, $\frac{1}{x} + \frac{1}{y} = \frac{7}{12}$

ANSWER $(3,4), (4,3)$

2. Solve $xy = 12$, $x^2 + y^2 = 25$

ANSWER $(\pm 3, \pm 4)$ matching signs

3. Solve $\frac{2}{x} - \frac{1}{y} = 1$, $\frac{1}{x} + \frac{2}{y} = 3$

ANSWER $(1, 1)$

4. Solve $x + y = 6$, $x^2 + y^2 = 20$

ANSWER $(2,4), (4,2)$

5. Solve $x - y = 2$, $\frac{1}{x} - \frac{1}{y} = -\frac{1}{4}$

ANSWER $(4,2)$ or $(-2,-4)$

6. Solve $xy = 6$, $x + y = -5$

ANSWER $(-2,-3), (-3,-2)$

7. How many solutions has $x + y = 1$, $xy = 1$?

ANSWER none — $D < 0$

Hard · stretch and reason

7 PROBLEMS

1. Solve $x + y = 4$, $x^3 + y^3 = 28$

ANSWER $(1,3), (3,1)$

2. Solve $x^2 + y^2 = 13$, $\frac{1}{x} + \frac{1}{y} = \frac{5}{6}$

ANSWER $(2,3), (3,2)$

3. Solve $\frac{x}{y} + \frac{y}{x} = \frac{5}{2}$, $x + y = 6$

ANSWER $(2,4), (4,2)$

4. Solve $x + y + xy = 11$, $x^2y + xy^2 = 30$

ANSWER $(2,3), (3,2)$ or $s = 6, p = 5$

5. Find k so $x + y = 4$, $xy = k$ has exactly one solution

ANSWER $k = 4$

6. Solve $\frac{1}{x} + \frac{1}{y} = \frac{1}{2}$, $\frac{1}{x^2} + \frac{1}{y^2} = \frac{5}{36}$

ANSWER $(6,3), (3,6)$

7. Interpret $x + y = 5$, $xy = 6$ graphically

ANSWER A line meeting a hyperbola in two points

07 Homework

Homework — 5 tasks

Write both ordered pairs whenever the system is symmetric.

1. Solve $x + y = 9$, $xy = 20$.
2. Solve $x + y = 5$, $\frac{1}{x} + \frac{1}{y} = \frac{5}{6}$.
3. Solve $x^2 + y^2 = 41$, $xy = 20$.

4. Solve $\frac{3}{x} + \frac{2}{y} = 4$, $\frac{1}{x} - \frac{1}{y} = 1$.

5. Explain, with a sketch, why $x + y = 1$ and $xy = 1$ have no common solution.

Rational inequalities

Never cross-multiply. Move everything to one side, combine into one fraction, and let the sign chart do the work.

LESSON 32–34

3 × 40 MIN

ALGEBRA 10, §2.3

P1 · 1.2 (EXTENSION)

LESSON CLOCK



The forbidden step 12 min

One fraction, compared with zero 20 min

The sign chart 23 min

Repeated factors 20 min

Practice 35 min

Homework 10 min

LEARNING OBJECTIVES

- Explain why multiplying an inequality by a variable expression is forbidden.
- Reduce a rational inequality to a single fraction compared with zero.
- Build and read a sign chart, including the excluded points.
- Write the answer in interval notation.

TEXTBOOK

National pp. 114–128

Cambridge Pure Mathematics 1, pp. 6–11

Workbook P1 Exercise 1C

01 Explanation

The one thing you may not do

NEVER MULTIPLY AN INEQUALITY BY $x - 3$

You do not know its sign. If it is negative the inequality reverses; if it is zero it collapses. $\frac{1}{x} > 1$ does **not** give $1 > x$ — try $x = -1$: $-1 > 1$ is false, but $1 > -1$ is true.

Multiplying by a **positive number** is safe. Multiplying by an expression whose sign you do not know is not, and there is no way to repair it afterwards.

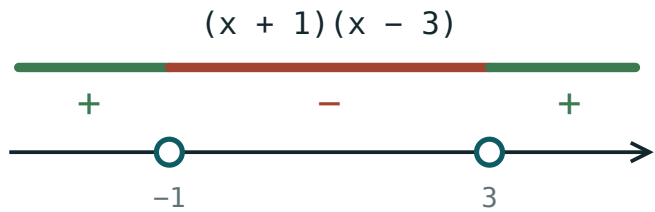
The method

FOUR STEPS, ALWAYS

1 Move everything to one side, so the other side is 0. **2** Combine into a single fraction and factorise top and bottom. **3** Mark every zero of the numerator and of the denominator on a number line. **4** Test the sign in each interval and read the answer off.

$$\frac{1}{x} > 1 \Rightarrow \frac{1}{x} - 1 > 0 \Rightarrow \frac{1-x}{x} > 0$$

Critical values 0 and 1. Testing -1, 0.5, 2 gives -, +, -, so the answer is $0 < x < 1$.



negative between the roots, positive outside

The critical values cut the line into intervals. One test point settles each.

Open or closed?

CRITICAL VALUE FROM	INEQUALITY	ENDPOINT
numerator	$\leq$ or $\geq$	included — closed circle
numerator	$<$ or $>$	excluded — open circle
denominator	any	always excluded

A DENOMINATOR ZERO IS NEVER INCLUDED

Even with $\geq$. The expression has no value there at all, so it cannot satisfy anything.

Repeated factors

The sign alternates as you cross a simple zero. It does **not** alternate at a repeated factor of even order — the graph touches the axis and turns back.

$$\frac{(x-1)^2(x+2)}{x-3} \geq 0$$

X	< -2	-2	$(-2,1)$	1	$(1,3)$	3	> 3
sign	+	0	-	0	-	x	+

Answer: $x \leq -2$, or $x = 1$, or $x > 3$. The isolated point 1 is part of the solution because the numerator is zero there and $\geq$ allows it.

A QUICK RULE

Start at the far right, where every factor is positive, and count the sign changes leftwards. A factor of odd power changes the sign; a factor of even power does not.

02 Worked examples

EXAMPLE 1 Solve $\frac{x-2}{x+3} < 0$.

- ① Already one fraction against zero.

- ② Critical values 2 and -3 .

- ③ Test $-4, 0, 3$: $+, -, +$.

- ④ Both ends open.

ANSWER

$$-3 < x < 2$$

EXAMPLE 2 Solve $\frac{2}{x-1} \leq 3$.

① $\frac{2}{x-1} - 3 \leq 0$

② $\frac{2-3(x-1)}{x-1} = \frac{5-3x}{x-1} \leq 0$

③ Critical $x = \frac{5}{3}$ (closed), $x = 1$ (open).

④ Test 0, 1.2, 2: -, +, -.

ANSWER

$$x < 1 \text{ or } x \geq \frac{5}{3}$$

EXAMPLE 3 Solve $\frac{(x+1)(x-4)}{x^2} > 0$.

① Critical $-1, 0, 4$.
 $x = 0$ is a double root of the denominator.

② Signs: +, -, -, + across the four intervals.
 x^2 never changes sign.

③ Ends all open.

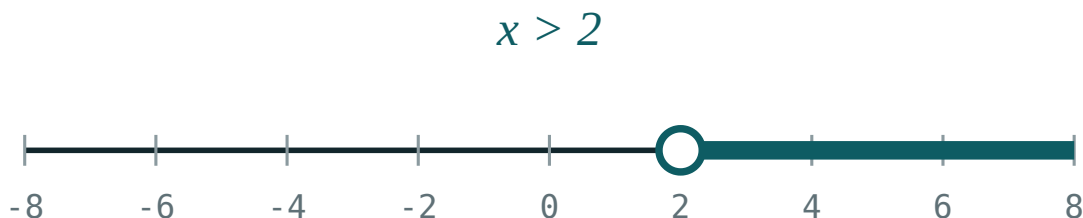
ANSWER

$$x < -1 \text{ or } x > 4$$

03 Interactive model

Draw the number line first, then fill in the signs interval by interval.

Reading a sign chart

inequality $x > 2$ boundary **2** (open) $x = 0$ works? **no** $x = 6$ works? **yes****04** Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Rational inequality	Ratsional tengsizlik	Рациональное неравенство
Sign chart (interval method)	Oraliqlar usuli	Метод интервалов
Critical value	Kritik qiymat	Критическое значение
Excluded point	Chetlatilgan nuqta	Выколота́я точка
Closed circle	To'ldirilgan nuqta	Закрашенная точка
Open circle	Bo'sh nuqta	Выколота́я точка
Interval notation	Oraliq belgilanishi	Интервальная запись
Union of intervals	Oraliqlar birlashmasi	Объединение промежутков
Sign of a factor	Ko'paytuvchi ishorasi	Знак множителя
Multiplicity	Karralilik	Кратность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

1 $x > 1$ gives:

A zero of the denominator is:

Crossing a factor of even power the sign:

$x - 2$ $x + 3 < 0$ gives:

$$x < -3$$

$$-3 < x < 2$$

$$x > 2$$

$$x < 2$$

The first step of a rational inequality is:

cross-multiply

get zero on one side

square it

take reciprocals

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\frac{1}{x} > 0$

ANSWER $x > 0$

2. Solve $\frac{1}{x} < 0$

ANSWER $x < 0$

3. Solve $\frac{x}{x-1} > 0$

ANSWER $x < 0$ or $x > 1$

4. Solve $\frac{x-3}{x+1} < 0$

ANSWER $-1 < x < 3$

5. Solve $(x-1)(x+2) > 0$

ANSWER $x < -2$ or $x > 1$

6. Is $x = 2$ allowed in $\frac{1}{x-2} \geq 0$?

ANSWER no

7. Solve $\frac{2}{x} \geq 0$

ANSWER $x > 0$

Medium · the standard question

7 PROBLEMS

1. Solve $\frac{1}{x} > 1$

ANSWER $0 < x < 1$

2. Solve $\frac{2}{x-1} \leq 3$

ANSWER $x < 1$ or $x \geq \frac{5}{3}$

3. Solve $\frac{x+2}{x-3} \geq 0$

ANSWER $x \leq -2$ or $x > 3$

4. Solve $\frac{x}{x^2-4} > 0$

ANSWER $-2 < x < 0$ or $x > 2$

5. Solve $\frac{(x+1)(x-4)}{x^2} > 0$

ANSWER $x < -1$ or $x > 4$

6. Solve $\frac{3}{x+2} < 1$

ANSWER $x < -2$ or $x > 1$

7. Solve $\frac{x-1}{x+1} \leq 2$

ANSWER $x < -1$ or $x \geq -3 \rightarrow x \leq -3$ or $x > -1$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{(x-1)^2(x+2)}{x-3} \geq 0$

ANSWER $x \leq -2, x = 1, \text{ or } x > 3$

2. Solve $\frac{x^2 - 5x + 6}{x^2 - 4} \leq 0$

ANSWER $-2 < x \leq 2$ excluding 2 $\rightarrow -2 < x < 2$ or $x = 3$

3. Solve $\frac{1}{x-1} > \frac{1}{x+1}$

ANSWER $x > 1$ or $-1 < x \rightarrow -1 < x < 1$ fails; $x > 1$ or $x < -1$

4. Solve $\frac{x}{x-2} \geq \frac{3}{x-2}$

ANSWER $x > 2$

5. Solve $\frac{x^3 - x}{x+3} < 0$

ANSWER $x < -3$ or $-1 < x < 0$ or $x > 1$ — check signs

6. Find all k with $\frac{x^2 + k}{x} > 0$ for every $x > 0$

ANSWER $k \geq 0$

7. Explain why $\frac{(x-2)^2}{x-2}$ and $x-2$ give different solution sets

ANSWER The first excludes $x = 2$

07 Homework

Homework — 6 tasks

Every answer needs the number line drawn, with open and closed circles marked.

1. Solve $\frac{x-5}{x+2} < 0$.

2. Solve $\frac{3}{x-2} \geq 1$.

3. Solve $\frac{x}{x^2-9} \leq 0$.

4. Solve $\frac{(x-2)^2(x+1)}{x-5} \geq 0$.

5. Explain in three sentences why $\frac{1}{x} > 1$ is not $1 > x$.

6. Solve $\frac{2}{x+1} < \frac{1}{x}$.

Systems of rational inequalities

Solve each one alone, draw them on the same line, and keep only the overlap.

LESSON 35–37

3 × 40 MIN

ALGEBRA 10, §2.4

P1 · 1.2 (EXTENSION)

LESSON CLOCK

9'

20'

23'

21'

34'

13'

And versus or	9 min
The method	20 min
Worked systems	23 min
Empty and unbounded answers	21 min
Practice	34 min
Homework	13 min

LEARNING OBJECTIVES

- Solve each inequality of a system separately.
- Represent the solution sets on one number line and read the intersection.
- Distinguish a system (“and”) from a collection (“or”).
- Handle systems whose solution set is empty.

TEXTBOOK

National	pp. 129–142
Cambridge	Pure Mathematics 1, pp. 6–11
Workbook	P1 Exercise 1C

01

Explanation

“And” or “or”

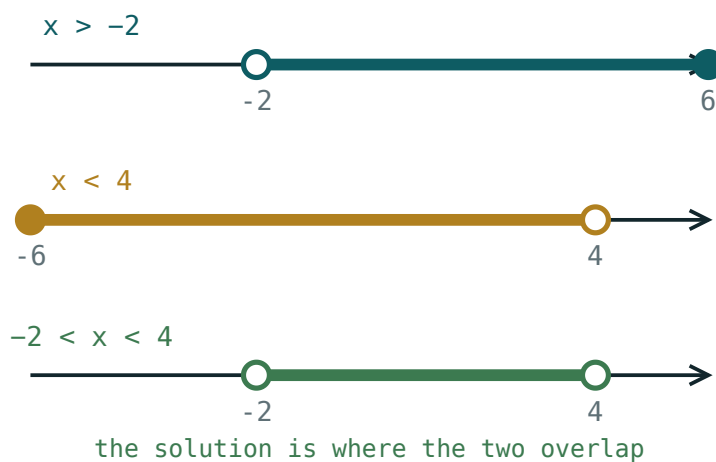
TWO DIFFERENT QUESTIONS

A **system** — written with a brace — asks for the values satisfying **every** inequality: the **intersection**. A **collection** asks for those satisfying **at least one**: the **union**.

$x > 1$ and $x < 4$ gives $1 < x < 4$. The same two as a collection give all of $\mathbb{R}$. The brace matters.

The method

1. Solve each inequality on its own, by the sign-chart method of the last lesson.
2. Draw all the solution sets on **one** number line, one above another.
3. Shade the part covered by every line.
4. Read the answer as an interval or a union of intervals.



Three solution sets drawn above one another. The answer is the strip covered by all three.

DO NOT TRY TO SOLVE THEM SIMULTANEOUSLY

Inequalities are not equations: adding or subtracting them loses information. Solve each separately, then intersect on the picture.

Worked systems

Example. $\frac{1}{x} > 0$ and $\frac{x-2}{x+1} \leq 0$.

INEQUALITY	SOLUTION
$\frac{1}{x} > 0$	$x > 0$
$\frac{x-2}{x+1} \leq 0$	$-1 < x \leq 2$
intersection	$0 < x \leq 2$

A double inequality such as $-3 < \frac{2}{x} < 1$ is a system in disguise: break it into $\frac{2}{x} >$

-3 and $\frac{2}{x} < 1$, solve both, intersect. It cannot be solved “all at once” because the middle contains x .

When the answer is empty

$x > 5$ and $x < 2$ have no common value. That is a legitimate answer, written $\emptyset$ or “no solution”, and it must be justified by the picture — not left blank.

THREE SHAPES OF ANSWER

a single interval · a union of intervals · the empty set. Anything else means an error in one of the individual solutions.

Systems are the natural home of the **domain** conditions from Quarter I. “Find the

domain of $\frac{\sqrt{x-1}}{x-4}$ ” is exactly the system $x-1 \geq 0$ and $x \neq 4$.

02

Worked examples

EXAMPLE 1 Solve the system $2x - 1 > 3$ and $\frac{1}{x-5} < 0$.

① First: $x > 2$.

② Second: $x < 5$.

The numerator is positive.

③ Intersect.

ANSWER

$$2 < x < 5$$

EXAMPLE 2 Solve $\frac{x}{x-3} \geq 0$ and $x^2 - 16 < 0$.

- 1 First: $x \leq 0$ or $x > 3$.
- 2 Second: $-4 < x < 4$.
- 3 Intersect both pieces.

ANSWER

$$-4 < x \leq 0 \text{ or } 3 < x < 4$$

EXAMPLE 3 Find the domain of $f(x) = \frac{\sqrt{x+2}}{x^2-9}$.

- 1 Radicand ≥ 0 : $x \geq -2$.
- 2 Denominator $\neq 0$: $x \neq \pm 3$.
- 3 $x = -3$ is already excluded by the first.

ANSWER

$$x \geq -2, x \neq 3$$

03 Interactive model

Draw both sets on the board one above the other; the answer is visible before it is written.

Intersecting two solution sets

$$x > 2$$

inequality $x > 2$ boundary **2** (open) $x = 0$ works? **no** $x = 6$ works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
System of inequalities	Tengsizliklar sistemasi	Система неравенств
Collection of inequalities	Tengsizliklar majmuasi	Совокупность неравенств
Intersection of sets	To'plamlar kesishmasi	Пересечение множеств
Union of sets	To'plamlar birlashmasi	Объединение множеств
Empty solution set	Bo'sh yechimlar to'plami	Пустое множество решений
Number line	Sonlar o'qi	Числовая прямая
Common part	Umumiy qism	Общая часть
Double inequality	Qo'sh tengsizlik	Двойное неравенство
Bounded interval	Chegaralangan oraliq	Ограниченный промежуток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A system of inequalities asks for:

the union

the intersection

either one

the difference

$x > 1$ and $x < 4$ gives:

$1 < x < 4$

$\mathbb{R}$

$x > 4$

$\emptyset$

$x > 5$ and $x < 2$ gives:

$2 < x < 5$

all reals

$\emptyset$

$x > 5$

$-3 < \underline{2} x < 1$ should be:

solved at once

split into two inequalities

cross-multiplied

squared

The domain of $\sqrt{x} - 1$ is:

$x \geq 0$

$x \geq 0, x \neq 1$

$x > 1$

$\mathbb{R}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $x > 2$ and $x < 7$

ANSWER $2 < x < 7$

2. Solve $x \geq 0$ and $x \leq 5$

ANSWER $0 \leq x \leq 5$

3. Solve $x > 4$ and $x < 1$

ANSWER $\emptyset$

4. Solve $2x > 6$ and $x - 1 < 4$

ANSWER $3 < x < 5$

5. Domain of $\frac{\sqrt{x}}{x-2}$

ANSWER $x \geq 0, x \neq 2$

6. Solve $x^2 < 9$

ANSWER $-3 < x < 3$

7. Solve $x > 0$ and $x^2 < 4$

ANSWER $0 < x < 2$

Medium · the standard question

7 PROBLEMS

1. Solve $2x - 1 > 3$ and $\frac{1}{x-5} < 0$

ANSWER $2 < x < 5$

2. Solve $\frac{x}{x-3} \geq 0$ and $x^2 - 16 < 0$

ANSWER $-4 < x \leq 0$ or $3 < x < 4$

3. Solve $-3 < \frac{2}{x} < 1$

ANSWER $x < -\frac{2}{3}$ or $x > 2$

4. Domain of $\frac{\sqrt{x+2}}{x^2-9}$

ANSWER $x \geq -2, x \neq 3$

5. Solve $\frac{1}{x} > 1$ and $x > 0.5$

ANSWER $0.5 < x < 1$

6. Solve $x^2 - 5x + 6 \leq 0$ and $x \neq 2$

ANSWER $2 < x \leq 3$

7. Solve $\frac{x-1}{x+2} > 0$ and $x < 3$

ANSWER $x < -2$ or $1 < x < 3$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{1}{x-1} > 2$ and $\frac{1}{x} < 3$

ANSWER $\frac{1}{3} < x < 1.5$ — check ends

2. Solve $x^2 - 4x < 0$ and $\frac{x-3}{x} \geq 0$

ANSWER $3 \leq x < 4$

3. Domain of $\frac{\sqrt{4-x^2}}{x-1}$

ANSWER $-2 \leq x \leq 2, x \neq 1$

4. Domain of $\sqrt{\frac{x-1}{x+2}}$

ANSWER $x < -2$ or $x \geq 1$

5. Find all k for which $x > k$ and $x < 2k$ is non-empty

ANSWER $k > 0$

6. Solve $|x-2| < 3$ and $\frac{1}{x} > 0$

ANSWER $0 < x < 5$

7. Solve $\frac{x^2-1}{x} \geq 0$ and $x \leq 2$

ANSWER $-1 \leq x < 0$ or $1 \leq x \leq 2$

07 Homework

Homework — 6 tasks

Draw one number line per system, with every solution set on it.

1. Solve $3x + 1 > 7$ and $\frac{1}{x-6} < 0$.

2. Solve $\frac{x+1}{x-2} \leq 0$ and $x > -3$.

3. Solve $-2 < \frac{3}{x} < 1$.

4. Find the domain of $\frac{\sqrt{x-1}}{x^2-16}$.

5. Find the domain of $\sqrt{\frac{x+3}{x-1}}$.

6. Give a system of two inequalities whose solution set is empty, and prove it with a number line.

Control work 3, and work on the mistakes

Rational equations, systems and inequalities in one paper — then the hour that turns the marks into a plan.

LESSON 38–39

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 3

P1 · CHAPTER 1 REVIEW

LESSON CLOCK

3

35'

9'

22'

11'

Instructions	3 min
The paper	35 min
Self-mark	9 min
Rewrite	22 min
Common-error drill	11 min

LEARNING OBJECTIVES

- Apply the rational-equation and rational-inequality methods under time.
- State a domain and reject an extraneous root without being prompted.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 143–146
Cambridge	Pure Mathematics 1, pp. 24–25
Workbook	Control paper C

01

Explanation

The paper — 25 marks, 40 minutes

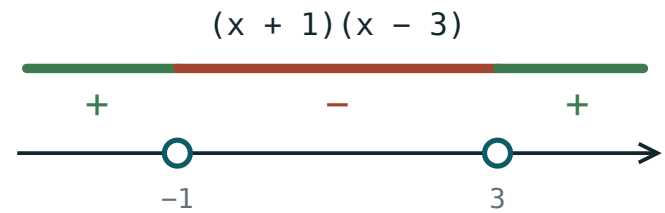
Q	TASK	MARKS	FROM
1	Solve $\frac{3}{x} + \frac{2}{x+1} = 3$, stating the domain	4	L28–29
2	Solve $\frac{x^2}{x-5} = \frac{25}{x-5}$ and justify the rejection	3	L28–29
3	Solve the system $x + y = 8$, $\frac{1}{x} + \frac{1}{y} = \frac{8}{15}$	4	L30–31
4	Solve $\frac{x-4}{x+2} \leq 0$ with a sign chart	4	L32–34
5	Solve $\frac{2}{x-3} > 1$	4	L32–34
6	Solve the system $\frac{x}{x-1} \geq 0$ and $x^2 < 9$	4	L35–37
7	Find the domain of $\frac{\sqrt{x+1}}{x^2-4}$	2	L35–37

WHERE THE METHOD MARKS ARE

Q1 and Q2 each give one mark for the domain line alone. Q4 and Q5 give two marks for the sign chart, whatever the final interval says. Working, not answers.

The four errors this topic produces

ERROR	LOOKS LIKE	KIND
no domain stated	an extraneous root kept	method
cross-multiplying an inequality	$\frac{1}{x} > 1 \Rightarrow 1 > x$	knowledge
denominator zero included	$x \leq 3$ where 3 kills the fraction	method
only one ordered pair	$(3, 5)$ without $(5, 3)$	careless



negative between the roots, positive outside

Questions 4, 5 and 6 in one picture. Two of the marks are for drawing it.

The rewrite

Every question that lost a mark is rewritten in full, with the domain on the first line and the check on the last. Under each rewrite, one sentence beginning “*I lost this because ...*”.

THE DRILL BEFORE THE REWRITE

Ten expressions on the board. For each, the class calls out only **the domain**, in three seconds. Nothing improves this topic faster.

02 Worked examples

EXAMPLE 1 Model answer, Q2: solve $\frac{x^2}{x-5} = \frac{25}{x-5}$.

- ① Domain $x \neq 5$.
One mark for this line.
- ② $x^2 = 25 \Rightarrow x = \pm 5$
- ③ $x = 5$ is excluded by the domain.
- ④ Only $x = -5$ survives.

ANSWER

$$x = -5$$

EXAMPLE 2 Model answer, Q5: solve $\frac{2}{x-3} > 1$.

① $\frac{2}{x-3} - 1 > 0$

Never cross-multiply.

② $\frac{2 - (x-3)}{x-3} = \frac{5-x}{x-3} > 0$

③ Critical 3 and 5, both open.

④ Test 0, 4, 6: $-$, $+$, $-$.

ANSWER

$$3 < x < 5$$

EXAMPLE 3 A learner wrote $\frac{x-4}{x+2} \leq 0 \Rightarrow -2 \leq x \leq 4$. Name the mistake.

① $x = -2$ makes the denominator zero.

② A denominator zero is never included.
Method error.

③ Correct: $-2 < x \leq 4$.

ANSWER

Method error — the excluded endpoint was closed

03 Interactive model

Run the domain drill before the rewrite, not after.

Mark these six answersDomain of $\frac{3}{x} + \frac{2}{x+1}$:

$x \neq 0$

$x \neq -1$

$x \neq 0, -1$

$\mathbb{R}$

 $\frac{x^2}{x-5} = \frac{25}{x-5}$ gives:

$x = \pm 5$

$x = 5$

$x = -5$

no solution

 $\frac{x-4}{x+2} \leq 0$ gives:

$-2 \leq x \leq 4$

$-2 < x \leq 4$

$-2 < x < 4$

$x \leq 4$

2 $x - 3 > 1$ gives:

The system $x + y = 8$, 1 x + 1 y = 8 15 gives:

Domain of $\sqrt{x+1} x^2 - 4$:

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Extraneous root	Chet ildiz	Посторонний корень
Sign chart	Ishoralar jadvali	Таблица знаков
Careless error	E'tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo'shlig'i	Пробел в знаниях
Mark scheme	Baholash sxemasi	Схема оценивания
Correction	Tuzatish	Исправление

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The domain line is worth:

nothing

a mark on its own

the whole question

half

Cross-multiplying an inequality is:

fine

fine if positive

never allowed with an unknown sign

always wrong

A denominator zero in a $\leq$ answer is:

included

excluded

sometimes included

the answer

A symmetric system is finished when you have:

one pair

both ordered pairs

the sum

the product

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $\frac{3}{x} + \frac{2}{x+1}$

ANSWER $x \neq 0, -1$

2. Solve $\frac{4}{x} = 2$

ANSWER $x = 2$

3. Solve $\frac{x-4}{x+2} \leq 0$

ANSWER $-2 < x \leq 4$

4. Solve $x^2 < 9$

ANSWER $-3 < x < 3$

5. Solve $x + y = 8, xy = 15$

ANSWER $(3,5), (5,3)$

6. Domain of $\sqrt{x+1}$

ANSWER $x \geq -1$

7. Solve $\frac{1}{x-3} > 0$

ANSWER $x > 3$

Medium · the standard question

7 PROBLEMS

1. Solve $\frac{3}{x} + \frac{2}{x+1} = 3$

ANSWER $x = \frac{3 \pm \sqrt{45}}{6}$ — i.e. $x \approx 1.62, -0.62$

2. Solve $\frac{x^2}{x-5} = \frac{25}{x-5}$

ANSWER $x = -5$

3. Solve $\frac{2}{x-3} > 1$

ANSWER $3 < x < 5$

4. Solve $\frac{x}{x-1} \geq 0$ and $x^2 < 9$

ANSWER $-3 < x \leq 0$ or $1 < x < 3$

5. Domain of $\frac{\sqrt{x+1}}{x^2-4}$

ANSWER $x \geq -1, x \neq 2$

6. Solve $x + y = 8, \frac{1}{x} + \frac{1}{y} = \frac{8}{15}$

ANSWER $(3,5), (5,3)$

7. Solve $\frac{x+3}{x-1} < 2$

ANSWER $x < 1$ or $x > 5$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{1}{x-2} + \frac{1}{x+2} = \frac{4}{x^2-4}$

ANSWER no solution — $x = 2$ rejected

2. Solve $\frac{x^2-9}{x^2-4x+3} \geq 0$

ANSWER $x < 1$ or $x \geq 3$, excluding $x = 3 \rightarrow x < 1$ or $x > 3$

3. Solve $x + y = 10, x^2 + y^2 = 58$

ANSWER $(3,7), (7,3)$

4. Domain of $\sqrt{\frac{x-2}{x+1}}$

ANSWER $x < -1$ or $x \geq 2$

5. Find k so $\frac{x}{x-k} = 2$ has no solution

ANSWER $k = 0$

6. Solve $\frac{1}{x} > \frac{1}{x-1}$

ANSWER $0 < x < 1$ fails; $x < 0$ or $x > 1$

7. Explain why the answer to Q4 is $-2 < x \leq 4$ and not $-2 \leq x \leq 4$

ANSWER The fraction is undefined at -2

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. It is not optional.

1. Rewrite in full every question that lost a mark, with the domain first and the check last.
2. Five problems from the section your knowledge column was heaviest in.

3. Solve $\frac{5}{x+2} \geq 1$ with a sign chart.
4. Solve $x + y = 12$, $xy = 32$, giving both ordered pairs.

Irrational equations

Squaring is the only way forward and the only source of false roots — so the check is not optional, it is the method.

LESSON 40–42

3 × 40 MIN

ALGEBRA 10, §2.5

P1 · 1.5 (EXTENSION)

LESSON CLOCK

12'

21'

27'

21'

33'

6'

Two conditions, not one	12 min
One radical	21 min
Two radicals	27 min
Substitutions	21 min
Practice	33 min
Homework	6 min

LEARNING OBJECTIVES

- State the two conditions an irrational equation imposes before squaring.
- Solve an equation with one radical by isolating and squaring.
- Solve an equation with two radicals by squaring twice.
- Check every candidate root in the original equation.

TEXTBOOK

National	pp. 147–162
Cambridge	Pure Mathematics 1, pp. 12–17
Workbook	P1 Exercise 1E

01

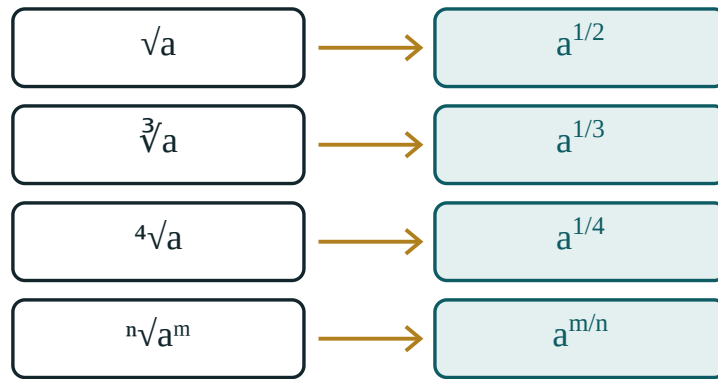
Explanation

Two conditions before you start

$\sqrt{A} = B$ REQUIRES BOTH

1 $A \geq 0$ — the radicand cannot be negative. **2** $B \geq 0$ — a square root is never negative, so the other side cannot be either. Condition 2 is the one that is forgotten, and it is the one that creates false roots.

$\sqrt{x + 3} = -2$ has no solution at all — the left side is never negative — but squaring gives $x + 3 = 4$ and the plausible-looking $x = 1$. Squaring destroys sign information, permanently.



a root is a power – the same object, two notations

The graph of a square root lives entirely above the axis. Nothing below it can be an answer.

One radical

1. **Isolate** the radical on one side, alone.
2. Square both sides.
3. Solve the polynomial equation.
4. **Check every root** in the *original* equation.

$$\sqrt{2x+3} = x \Rightarrow 2x+3 = x^2 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow x = 3, -1$$

Check: $x = 3$ gives $\sqrt{9} = 3$ ✓. $x = -1$ gives $\sqrt{1} = -1$ ✗. Only $x = 3$.

ISOLATE FIRST, SQUARE SECOND

Squaring $\sqrt{x} + 1 = 4$ directly gives $x + 2\sqrt{x} + 1 = 16$ — a new radical, and more work. Move the 1 across first.

Two radicals

Arrange one radical on each side, square, then isolate whatever radical survives and square again:

$$\sqrt{x+5} - \sqrt{x} = 1 \Rightarrow \sqrt{x+5} = 1 + \sqrt{x}$$

$$x+5 = 1 + 2\sqrt{x} + x \Rightarrow 4 = 2\sqrt{x} \Rightarrow \sqrt{x} = 2 \Rightarrow x = 4$$

Check: $\sqrt{9} - \sqrt{4} = 3 - 2 = 1$ ✓.

WHY ONE ON EACH SIDE

$(a - b)^2$ still contains a cross term with both radicals; $(1 + \sqrt{x})^2$ contains only one. Arranging the equation before squaring halves the work.

Substitutions

When the same radical appears more than once, name it:

$$x + \sqrt{x} - 6 = 0, \text{ let } t = \sqrt{x} \geq 0$$

Then $t^2 + t - 6 = 0$, so $t = 2$ or $t = -3$. The second is impossible because $t \geq 0$, so $\sqrt{x} = 2$ and $x = 4$.

The condition $t \geq 0$ is written down with the substitution, not remembered later. It is what removes the false root before any checking is needed.

02 Worked examples

EXAMPLE 1 Solve $\sqrt{3x + 1} = x - 1$.

① Conditions: $3x + 1 \geq 0$ and $x - 1 \geq 0$, so $x \geq 1$.

② $3x + 1 = x^2 - 2x + 1$

③ $x^2 - 5x = 0 \Rightarrow x = 0, 5$

④ $x = 0$ fails $x \geq 1$.

ANSWER

$$x = 5$$

EXAMPLE 2 Solve $\sqrt{x+7} - \sqrt{x} = 1$.

① $\sqrt{x+7} = 1 + \sqrt{x}$
One on each side.

② $x + 7 = 1 + 2\sqrt{x} + x$

③ $6 = 2\sqrt{x} \Rightarrow \sqrt{x} = 3$

④ $x = 9$. Check: $4 - 3 = 1$ ✓

ANSWER

$x = 9$

EXAMPLE 3 Solve $x - 5\sqrt{x} + 6 = 0$.

① Let $t = \sqrt{x} \geq 0$.
 $t^2 - 5t + 6 = 0$

② $t = 2$ or $t = 3$, both allowed.

③ $\sqrt{x} = 2 \Rightarrow x = 4$

④ $\sqrt{x} = 3 \Rightarrow x = 9$

ANSWER

$x = 4$ and $x = 9$

03 Interactive model

Sketch both sides as graphs — the intersections are the true roots, and false roots are visibly absent.

Roots and powers

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value³ **64**

a¹ **64**

Raising the answer to the power 3 gives back a¹ — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Irrational equation	Irratsional tenglama	Иррациональное уравнение
Radical	Radikal	Радикал
Radicand	Ildiz ostidagi ifoda	Подкоренное выражение
Isolate the radical	Radikalni ajratish	Уединить радикал
Squaring both sides	Kvadratga ko‘tarish	Возведение в квадрат
False (extraneous) root	Chet ildiz	Посторонний корень
Necessary condition	Zaruriy shart	Необходимое условие
Equivalent transformation	Teng kuchli almashtirish	Равносильное преобразование
Verification	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sqrt{A} = B$ requires:

$A \geq 0$

$B \geq 0$

both

neither

$\sqrt{x+3} = -2$ has:

$x = 1$

$x = -7$

no solution

two solutions

The first step with $\sqrt{x} + 1 = 4$ is:

square

isolate the radical

substitute

cube

For $x + \sqrt{x} - 6 = 0$ set:

$$t = x$$

$$t = \sqrt{x}$$

$$t = x^2$$

nothing

After squaring you must:

stop

check each root in the original

square again

factorise

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sqrt{x} = 3$

ANSWER $x = 9$

2. Solve $\sqrt{x+1} = 2$

ANSWER $x = 3$

3. Solve $\sqrt{2x} = 4$

ANSWER $x = 8$

4. Solve $\sqrt{x} = -1$

ANSWER no solution

5. Solve $\sqrt{x-3} = 0$

ANSWER $x = 3$

6. Solve $\sqrt{5-x} = 1$

ANSWER $x = 4$

7. Condition on x in $\sqrt{x-2}$

ANSWER $x \geq 2$

Medium · the standard question

7 PROBLEMS

1. Solve $\sqrt{2x+3} = x$

ANSWER $x = 3$

2. Solve $\sqrt{3x+1} = x-1$

ANSWER $x = 5$

3. Solve $\sqrt{x+7} - \sqrt{x} = 1$

ANSWER $x = 9$

4. Solve $x - 5\sqrt{x} + 6 = 0$

ANSWER $x = 4, 9$

5. Solve $\sqrt{x+4} = x-2$

ANSWER $x = 5$

6. Solve $\sqrt{x^2-3} = 1$

ANSWER $x = \pm 2$

7. Solve $\sqrt{x} + 1 = 4$

ANSWER $x = 9$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sqrt{x+5} + \sqrt{x} = 5$

ANSWER $x = 4$

2. Solve $\sqrt{2x+5} - \sqrt{x-1} = 2$

ANSWER $x = 10$ or $x = 2$

3. Solve $x + \sqrt{x+5} = 7$

ANSWER $x = 4$

4. Solve $\sqrt{x^2 + 3x} = x + 1$

ANSWER $x = 1$

5. Solve $2x - 7\sqrt{x} + 3 = 0$

ANSWER $x = 9$ or $x = 0.25$

6. Solve $\sqrt{\sqrt{x}} = 2$

ANSWER $x = 16$

7. Find all k for which $\sqrt{x} = x + k$ has exactly one solution

ANSWER $k = 0.25$ or $k \leq 0$

07 Homework

Homework — 6 tasks

Every answer must show the check written out in full.

1. Solve $\sqrt{4x+1} = x - 1$.

2. Solve $\sqrt{x+9} - \sqrt{x} = 1$.

3. Solve $x - 7\sqrt{x} + 12 = 0$.

4. Solve $\sqrt{2x-1} = x - 2$.

5. Explain, in three sentences, why squaring can create a root the original equation never had.
6. Show that $\sqrt{x+2} = -3$ has no solution without squaring.

Systems of irrational equations

Four lessons on the systems that hide a quadratic — sum and difference of roots, substitution, and the conditions that must survive to the end.

LESSON 43–46

4 × 40 MIN

ALGEBRA 10, §2.6

P1 · 1.5–1.7 (EXTENSION)

LESSON CLOCK

14	27'	27'	27'	44'	21'
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Where the difficulty is 14 min

$u = \sqrt{x}$, $v = \sqrt{y}$ 27 min

Sum and product of roots 27 min

Mixed systems 27 min

Practice 44 min

Homework and consolidation 21 min

LEARNING OBJECTIVES

- Solve a system of irrational equations by substitution.
- Use $u = \sqrt{x}$, $v = \sqrt{y}$ to turn a system into a polynomial one.
- Track the non-negativity conditions through every step.
- Check every candidate pair in the original system.

TEXTBOOK

National pp. 163–180

Cambridge Pure Mathematics 1, pp. 12–23

Workbook P1 Exercise 1E, 1F

01 Explanation

Where the difficulty is

Nothing new is being asked. A system of irrational equations combines two techniques you already have — the substitution of the last lesson and the elimination of Lesson 30–31. What is new is the **bookkeeping**: two non-negativity conditions instead of one, and they must survive every step.

WRITE THE CONDITIONS ONCE, AT THE TOP

$x \geq 0$, $y \geq 0$, and any others the radicands impose. Then substitute $u = \sqrt{x} \geq 0$ and $v = \sqrt{y} \geq 0$ — and every rejection later is decided by those two inequalities alone.

The standard substitution

$$\sqrt{x} + \sqrt{y} = 5 \text{ and } x + y = 13$$

Put $u = \sqrt{x}$, $v = \sqrt{y}$, both ≥ 0 . Then $x = u^2$ and $y = v^2$, and the system becomes polynomial:

$$u + v = 5 \text{ and } u^2 + v^2 = 13$$

So $uv = \frac{25 - 13}{2} = 6$, and u, v are the roots of $t^2 - 5t + 6 = 0$: 2 and 3. Both are non-negative, so both are kept.

$(x, y) = (4, 9) \text{ or } (9, 4)$

SQUARE THE SUBSTITUTION BACK CAREFULLY

$u = 2$ gives $x = 4$, not $x = 2$. Half the errors in this lesson are made in the last line.

Sum and product of the roots

Many systems are stated directly in $\sqrt{x} + \sqrt{y}$ and $\sqrt{xy}$. Since $\sqrt{xy} = \sqrt{x} \cdot \sqrt{y}$ for non-negative x, y , these are exactly $u + v$ and uv :

GIVEN	IN u, v
$\sqrt{x} + \sqrt{y}$	$u + v = s$
$\sqrt{xy}$	$uv = p$
$x + y$	$s^2 - 2p$
$x - y$	$(u - v)(u + v)$

Then u and v are the roots of $t^2 - st + p = 0$, and both must come out non-negative or the pair is rejected.

Mixed systems

When only one equation carries a radical, isolate and substitute in the ordinary way:

$\sqrt{x + y} = 3 \text{ and } x - y = 1$

The first gives $x + y = 9$ (and requires $x + y \geq 0$, which it now is), so the system is linear: $x = 5, y = 4$. Check: $\sqrt{9} = 3$ ✓.

THE FINAL CHECK IS ON THE ORIGINAL SYSTEM

Not on the substituted one. A pair that satisfies $u + v = 5$ may still fail $\sqrt{x} + \sqrt{y} = 5$ if a sign was lost. Substitute the numbers back into the equations as first written.

02 Worked examples

EXAMPLE 1 Solve $\sqrt{x} + \sqrt{y} = 7, x + y = 25$.

① $u + v = 7, u^2 + v^2 = 25, \text{ both } \geq 0.$

② $uv = \frac{49 - 25}{2} = 12$

③ $t^2 - 7t + 12 = 0 \Rightarrow t = 3, 4$

④ $x = 9, y = 16$ or the reverse.

ANSWER

$(9, 16)$ and $(16, 9)$

EXAMPLE 2 Solve $\sqrt{x} - \sqrt{y} = 1, x + y = 25$.

① $u - v = 1, u^2 + v^2 = 25$.

② $u = v + 1 \Rightarrow (v+1)^2 + v^2 = 25$

③ $2v^2 + 2v - 24 = 0 \Rightarrow v^2 + v - 12 = 0$
 $v = 3$ or $v = -4$ (rejected)

④ $v = 3, u = 4$

ANSWER

(16, 9)

EXAMPLE 3 Solve $\sqrt{x+y} = 4, x - y = 2$.

① $x + y = 16$
 Condition satisfied.

② $x - y = 2$

③ Add: $2x = 18 \Rightarrow x = 9$.

④ $y = 7$. Check $\sqrt{16} = 4$ ✓

ANSWER

(9, 7)

03 Interactive model

Write the two conditions on the board and keep them visible for the whole solution.

Substituting a root

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value³ **64**

a¹ **64**

Raising the answer to the power 3 gives back a¹ — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
System of irrational equations	Irratsional tenglamalar sistemasi	Система иррациональных уравнений
Non-negativity condition	Manfiy emaslik sharti	Условие неотрицательности
Auxiliary variable	Yordamchi o‘zgaruvchi	Вспомогательная переменная
Sum of roots	Ildizlar yig‘indisi	Сумма корней
Product of roots	Ildizlar ko‘paytmasi	Произведение корней
Back-substitution	Teskari almashtirish	Обратная подстановка
Rejected pair	Rad etilgan juftlik	Отброшенная пара
Symmetric system	Simmetrik sistema	Симметричная система
Verification	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In $u = \sqrt{x}$ the condition is:

If $u = 3$ then x is:

$\sqrt{x} + \sqrt{y} = 5$, $x + y = 13$ gives uv :

A negative value of v means:

a second answer

the pair is rejected

square it

nothing

The final check is done on:

the substituted system

the original system

neither

the quadratic

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sqrt{x} = 2, y = x + 1$

ANSWER (4, 5)

2. Solve $\sqrt{x + y} = 3, x = y$

ANSWER (4.5, 4.5)

3. If $u = \sqrt{x} = 5$, find x

ANSWER 25

4. Condition on $u = \sqrt{x}$

ANSWER $u \geq 0$

5. Solve $\sqrt{x} + \sqrt{y} = 0$

ANSWER (0, 0)

6. Solve $\sqrt{x} = 3, \sqrt{y} = 4$

ANSWER (9, 16)

7. Write $x + y$ in terms of u, v

ANSWER $u^2 + v^2$

Medium · the standard question

7 PROBLEMS

1. Solve $\sqrt{x} + \sqrt{y} = 5, x + y = 13$

ANSWER (4,9), (9,4)

2. Solve $\sqrt{x} + \sqrt{y} = 7, x + y = 25$

ANSWER (9,16), (16,9)

3. Solve $\sqrt{x} - \sqrt{y} = 1, x + y = 25$

ANSWER (16, 9)

4. Solve $\sqrt{x+y} = 4, x - y = 2$

ANSWER (9, 7)

5. Solve $\sqrt{x} + \sqrt{y} = 6, \sqrt{xy} = 8$

ANSWER (4,16), (16,4)

6. Solve $\sqrt{x} \cdot \sqrt{y} = 6, x + y = 13$

ANSWER (4,9), (9,4)

7. Solve $\sqrt{x-1} = 2, x + y = 10$

ANSWER (5, 5)

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sqrt{x} + \sqrt{y} = 4, x + y = 10$

ANSWER $(3 \pm 2\sqrt{2})^2$ — i.e. $u, v = 2 \pm \sqrt{2}$

2. Solve $x + y = 20, \sqrt{x} - \sqrt{y} = 2$

ANSWER $(16, 4)$

3. Solve $\sqrt{x+y} + \sqrt{x-y} = 4, x^2 - y^2 = 9$

ANSWER $(5, 4)$

4. Solve $\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}} = \frac{5}{6}, xy = 36$

ANSWER $(4, 9), (9, 4)$

5. Solve $\sqrt{x} + \sqrt{y} = \sqrt{x+y}$

ANSWER $x = 0$ or $y = 0$

6. Explain why $\sqrt{x} + \sqrt{y} = 3, x + y = 10$ has no solution

ANSWER $uv = -0.5 < 0$ is impossible for $u, v \geq 0$

7. Solve $\sqrt{x} + \sqrt{y} = 5, x - y = 5$

ANSWER $(9, 4)$

07 Homework

Homework — 6 tasks

Write $u \geq 0$ and $v \geq 0$ on the first line of every solution.

1. Solve $\sqrt{x} + \sqrt{y} = 9, x + y = 45$.

2. Solve $\sqrt{x} - \sqrt{y} = 2, x + y = 20$.

3. Solve $\sqrt{x+y} = 5, x - y = 3$.

4. Solve $\sqrt{x} + \sqrt{y} = 8, \sqrt{xy} = 15$.
5. Explain why $\sqrt{x} + \sqrt{y} = 2, x + y = 10$ has no solution.
6. Check both answers of task 1 by substituting into the original system.

Control work 4, and the quarter review

Irrational equations and systems in one paper, then the map that links a domain to a check to an answer.

LESSON 47–48

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 4

P1 · CHAPTER 1 REVIEW

LESSON CLOCK

3

37'

9'

18'

9'

4'

Instructions	3 min
The paper	37 min
Answers	9 min
Rewrite	18 min
Concept map	9 min
Targets	4 min

LEARNING OBJECTIVES

- Apply every method of Quarter II in one assessment.
- Decide which technique a question wants without being told.
- Build a concept map linking domain, transformation and check.
- Set a personal target for Quarter III.

TEXTBOOK

National	pp. 181–184
Cambridge	Pure Mathematics 1, pp. 24–25
Workbook	Control paper D

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Solve $\sqrt{2x+7} = x+2$, with the conditions stated	5	L40–42
2	Solve $\sqrt{x+5} - \sqrt{x} = 1$	5	L40–42
3	Solve $x - 6\sqrt{x} + 8 = 0$ by substitution	4	L40–42
4	Solve $\sqrt{x} + \sqrt{y} = 6, x + y = 20$	6	L43–46
5	Solve $\frac{2}{x-1} \geq 1$ with a sign chart	5	L32–34
6	Find the domain of $\frac{\sqrt{x-2}}{x^2-25}$	5	L35–37

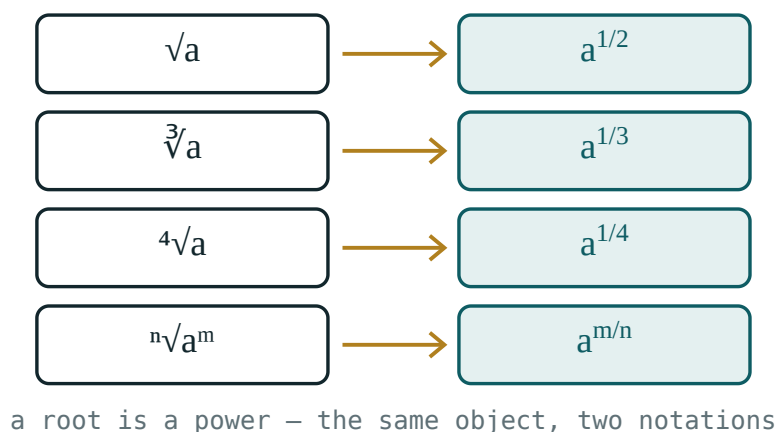
TWO MARKS ARE FOR CONDITIONS

Q1 gives one for $x + 2 \geq 0$ and one for the rejection that follows from it. A correct answer with neither scores three of five.

The concept map

Six boxes, links as sentences:

- **rational equation** → **domain** — “each denominator forbids one value”
- **domain** → **extraneous root** — “multiplying by zero is not reversible”
- **rational inequality** → **sign chart** — “never cross-multiply”
- **system** → **intersection** — “draw them on one line”
- **irrational equation** → **two conditions** — “radicand ≥ 0 and the other side ≥ 0 ”
- **squaring** → **check** — “sign information is destroyed, so it must be restored”



*The single picture behind the whole quarter: a root
never goes below the axis.*

Looking forward

Quarter III opens with the exponential and logarithmic functions. Their equations are solved by exactly the technique of this quarter: state the domain, transform, check. A logarithm demands *argument* > 0 before anything else — the same first line as $\sqrt{A}$ demands $A \geq 0$.

ONE HABIT TO CARRY FORWARD

The first line of every solution states what the unknown is allowed to be. Learners who built that habit this quarter will find Quarter III half-solved before they start.

02 Worked examples

EXAMPLE 1 Model answer, Q1: solve $\sqrt{2x+7} = x+2$.

- ① Conditions $2x+7 \geq 0$ and $x+2 \geq 0$, so $x \geq -2$.
- ② $2x+7 = x^2+4x+4$
- ③ $x^2+2x-3 = 0 \Rightarrow x = 1, -3$
- ④ $x = -3$ fails $x \geq -2$.

ANSWER

$$x = 1$$

EXAMPLE 2 Model answer, Q4: $\sqrt{x} + \sqrt{y} = 6$, $x+y = 20$.

- ① $u+v = 6$, $u^2+v^2 = 20$, both ≥ 0 .
- ② $uv = \frac{36-20}{2} = 8$
- ③ $t^2-6t+8 = 0 \Rightarrow t = 2, 4$
- ④ $x = 4, y = 16$ or the reverse.

ANSWER

$(4, 16)$ and $(16, 4)$

EXAMPLE 3 Model answer, Q6: domain of $\frac{\sqrt{x-2}}{x^2-25}$.

① $x - 2 \geq 0 \Rightarrow x \geq 2$

② $x^2 - 25 \neq 0 \Rightarrow x \neq \pm 5$

③ $x = -5$ is already excluded.

ANSWER

$x \geq 2, x \neq 5$

03 Interactive model

Work Q1 and Q4 on the board with the conditions written in a box at the top.

The quarter in ten questions

Domain of $\frac{1}{x-4} + \frac{1}{x}$:

$\frac{x^2}{x-3} = \frac{9}{x-3}$ gives:

$\frac{1}{x} > 2$ gives:

$x - 4$ $x + 2 \leq 0$ gives:

$$-2 \leq x \leq 4$$

$$-2 < x \leq 4$$

$$-2 < x < 4$$

$$x \leq 4$$

$\sqrt{2x + 7} = x + 2$ gives:

$$x = 1$$

$$x = 1, -3$$

$$x = -3$$

none

$\sqrt{x + 5} - \sqrt{x} = 1$ gives:

$$x = 4$$

$$x = 5$$

$$x = 9$$

$$x = 1$$

$x - 6\sqrt{x} + 8 = 0$ gives:

$x = 2, 4$

$x = 4, 16$

$x = 16$

$x = 8$

$\sqrt{x} + \sqrt{y} = 6, x + y = 20$ gives:

$(4,16)$ only

$(4,16), (16,4)$

$(2,4)$

none

$\frac{2}{x-1} \geq 1$ gives:

$x \geq 3$

$1 < x \leq 3$

$x \leq 3$

$x > 1$

Domain of $\sqrt{x-2} \cdot x^2 - 25$:

$x \geq 2$

$x \geq 2, x \neq 5$

$x \neq \pm 5$

$x > 2$

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Equivalent transformation	Teng kuchli almashtirish	Равносильное преобразование
Non-equivalent step	Teng kuchli bo‘lmagan qadam	Неравносильный переход
Extraneous root	Chet ildiz	Посторонний корень
Sign chart	Ishoralar jadvali	Таблица знаков
Target	Maqsad	Цель
Self-assessment	O‘z-o‘zini baholash	Самооценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first line of any solution this quarter states:

the answer

what the unknown may be

the method

the check

Squaring is:

always equivalent

equivalent only when both sides are ≥ 0

never allowed

reversible

Quarter III begins with:

trigonometry

the exponential function

probability

vectors

A logarithm will demand:

argument ≥ 0

argument > 0

base = 10

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sqrt{x} = 6$

ANSWER $x = 36$

2. Domain of $\frac{1}{x-4}$

ANSWER $x \neq 4$

3. Solve $\frac{3}{x} = 1$

ANSWER $x = 3$

4. Solve $\frac{x-4}{x+2} \leq 0$

ANSWER $-2 < x \leq 4$

5. Solve $x + y = 6, xy = 8$

ANSWER $(2,4), (4,2)$

6. Domain of $\sqrt{x-2}$

ANSWER $x \geq 2$

7. Solve $\sqrt{x+1} = 3$

ANSWER $x = 8$

Medium · the standard question

7 PROBLEMS

1. Solve $\sqrt{2x+7} = x+2$

ANSWER $x = 1$

2. Solve $\sqrt{x+5} - \sqrt{x} = 1$

ANSWER $x = 4$

3. Solve $x - 6\sqrt{x} + 8 = 0$

ANSWER $x = 4, 16$

4. Solve $\sqrt{x} + \sqrt{y} = 6, x + y = 20$

ANSWER $(4,16), (16,4)$

5. Solve $\frac{2}{x-1} \geq 1$

ANSWER $1 < x \leq 3$

6. Domain of $\frac{\sqrt{x-2}}{x^2-25}$

ANSWER $x \geq 2, x \neq 5$

7. Solve $\frac{x}{x+3} = \frac{2}{x}$

ANSWER $x = 6, -1$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sqrt{3x+4} - \sqrt{x+1} = 1$

ANSWER $x = 0$ or $x = 8$

2. Solve $\frac{x^2-1}{x^2-4} \leq 0$

ANSWER $-2 < x \leq -1$ or $1 \leq x < 2$

3. Solve $\sqrt{x} + \sqrt{y} = 5, xy = 36$

ANSWER $(4,9), (9,4)$

4. Domain of $\sqrt{\frac{x-3}{x+2}}$

ANSWER $x < -2$ or $x \geq 3$

5. Solve $2x + \sqrt{x} - 1 = 0$

ANSWER $x = 0.25$

6. Find k so $\sqrt{x} = x - k$ has exactly two solutions

ANSWER $0 < k < 0.25$

7. Solve $\frac{1}{\sqrt{x}} + \sqrt{x} = 2.5$

ANSWER $x = 4$ or $x = 0.25$

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter III.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.
3. Solve $\sqrt{4x+5} = x+2$ and $\sqrt{x+3} + \sqrt{x} = 3$.

4. Write your target for Quarter III in one checkable sentence, and date it.